Lab 3: Architectural Justification

Digital Gift Card & Loyalty Point Exchange

Architecture Selection

We chose Microservices Architecture for the Digital Gift Card & Loyalty Point Exchange platform.

Reason 1: Independent Scaling Under Fluctuating Load

The platform's requirements demand very different scaling profiles across functions. Voucher generation and merchant-side validation must sustain sub-250ms response times under peak load (NFR-001), while wallet aggregation and merchant account linking are comparatively low-frequency. Microservices let each capability — Wallet Aggregation, Point Conversion, Voucher Generation, and Merchant Validation — scale independently based on its own demand curve, rather than over-provisioning the entire system to protect one high-traffic path, as a monolithic layered design would require.

Reason 2: Fault Isolation Across Distinct Business Domains

The system spans several loosely related concerns: merchant ledger integration, point-to-credit conversion, voucher issuance, and merchant-side redemption. These map naturally to independently deployable services with clear domain boundaries. If the Merchant Ledger Service experiences a slowdown or outage while validating an external account, it should not block or crash the Voucher Generation or Merchant Validation services. A microservices boundary contains that failure, which a tightly-coupled layered or centralized client-server design would not.

Security Advantage

Isolating voucher checksum generation and validation inside a dedicated Security & Encryption Module (rather than embedding cryptographic logic inside each service) creates a single, auditable boundary for enforcing NFR-002 (TLS 1.2+ in transit, AES-256 at rest). Both the Voucher Generation Service and Merchant Validation Service consume this module through the same ISecurityModule interface, so encryption and checksum logic is implemented once, consistently, and can be independently hardened, patched, or audited without touching the services that depend on it — directly supporting the anti-fraud voucher locking requirement (FR-004).

Performance Benefit

Because each service owns a narrow responsibility, the Voucher Generation Service and Merchant Validation Service can be optimized and scaled specifically for the sub-250ms redemption and validation path (NFR-001), independent of slower, less time-sensitive operations like loyalty program management. This targeted scaling — rather than scaling an entire monolith to protect one latency-critical flow — makes it realistic to meet the 250ms threshold for 95% of transactions even under peak load, while keeping infrastructure cost proportional to actual demand.